Hasan Mohd Hussain

Software Development Engineer | *Backend Software Engineer* | *Python, REST APIs & AWS*
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PROFESSIONAL SUMMARY

Backend Software Engineer with an **MS in Computer Engineering**, building scalable **Python** backend services and **REST APIs** on **AWS**. Cut QA runtime **95%** through **test automation** and sped up **API responses 20%**. Skilled in **FastAPI, Flask, SQL/NoSQL, Docker, CI/CD** and **system design**.

EDUCATION

University of Texas at Dallas — **Richardson, Texas, USA**
Master of Science in Computer Engineering — Aug 2024 - Aug 2026
GPA: 3.74/4.0 | *Relevant Coursework: NLP, Machine Learning, Computer Architecture, Embedded Systems*

University of Mumbai — **Mumbai, Maharashtra, India**
Bachelor of Engineering in Computer Engineering — Aug 2017 - Jul 2021
GPA: 7.88/10.0 | *Relevant Coursework: Data Structures, Database Management Systems*

SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, SQL, Go, Bash
Backend & APIs: FastAPI, Flask, Django, Spring Boot, Node.js, Express, RESTful API design, microservices, distributed systems, GraphQL, gRPC, WebSockets, authentication (JWT, OAuth 2.0), caching, rate limiting, API security
Databases & Frontend: PostgreSQL, MySQL, MongoDB, SQLite, Redis, schema design, indexing, React, Next.js, HTML5, CSS3
Cloud & DevOps: AWS (ECS Fargate, ECR, EC2, S3), Docker, CI/CD, GitHub Actions, Git, Linux, CloudWatch, basic K8s/Terraform
Engineering & Testing: OOP, data structures & algorithms, system design, design patterns, SDLC, Agile/Scrum, code reviews, debugging, technical documentation, PyTest, unit testing, integration testing, TDD
AI Integration: LLM APIs, RAG, LangChain, LangGraph, OpenAI API, Azure OpenAI, Gemini API, Ollama, Claude Code, Claude Cowork

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer — Jun 2026 - Aug 2026
- Engineered an **end-to-end QA automation agent** in **Python, Flask** and **Playwright** for a **production mortgage dashboard**.
- Integrated **multi-provider LLM grading** (Claude, GPT-4o, Gemini, Qwen) with **PDF/Excel reports**, cutting QA runtime by **95%**.
- Developed a **synthetic test-data generator** for **5 financial document types**, scaling **regression test coverage** to **200 cases/run**.

Mphasis — **New York, United States**
Machine Learning Engineer — May 2025 - Aug 2025
- Delivered a **PDF RAG microservice** with **FastAPI REST endpoints**, **ChromaDB** and **Azure OpenAI**, improving retrieval by **30%**.
- Optimized **Python backend logic** for **FastAPI, Docker** and **AWS** services, improving **API response time by 20%**.
- Automated **loan-document PDF processing** (reordering, rotation, **JSON bookmark re-indexing**) in **Python/PyPDF2** for **banking clients**.

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Built **Unity/C#** gameplay systems from design docs, raising **user engagement 25%** at a stable **60 FPS**, using **Git**.

Shree Infotech — **Mumbai, India**
Software Engineer — Apr 2023 - Mar 2024
- Developed **Python/Flask backend services** and **REST APIs** on **Docker**, automating workflows and saving **10+ hours/week**.
- Designed **MongoDB** schemas and indexes for **fast query retrieval**, and wrote **unit tests** (PyTest) to keep core services reliable.

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Shipped a **2D physics-based driving game** in **Unity/C#** in **two weeks**: **500K downloads** in month one, **30%** higher engagement.

PROJECTS

JobQuest.AI | *Python, FastAPI, SQLite, Pydantic, JavaScript, REST APIs, Multi-LLM*
- Built a **full-stack job-search app** with **91 FastAPI REST endpoints**, **12 SQLite tables** and a **10-provider LLM layer** with **fallback**.
- Shipped **application tracking**, **Chart.js analytics** and **Gmail SMTP** outreach with **195 automated tests**, packaged via **PyInstaller**.

Recipe Manager | *Python, Flask, MongoDB, Docker, AWS ECR, ECS Fargate*
- Implemented **authenticated CRUD workflows** in **Flask/MongoDB** across **9 routes** (**6 login-protected**) with per-user data scoping.
- **Containerized** the app with **Docker/Gunicorn**, pushed to **Amazon ECR** and deployed on **AWS ECS Fargate** with **Secrets Manager**.

IronPay LedgerAPI | *Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Docker, pytest, k6*
- Designed a **UPI-style payments API**: **double-entry ledger**, **idempotent transfers**, **row-level locks** and a **6-table PostgreSQL schema**.
- Designing **JWT auth**, **Redis rate limiting**, a **transactional outbox** and **double-spend tests**, targeting **p95 < 300 ms** under **k6** load.

CERTIFICATIONS

Anthropic: Claude Code in Action, Claude Code 101, Claude 101 | **Udemy:** SQL Bootcamp | **DeepLearning.AI:** AI For Everyone
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows | **Meta:** Back-End Developer Professional Certificate